



Computer Systems Security

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Definitions

- An encryption scheme is defined by three algorithms: Gen , Enc , and Dec as well as a specification of a (finite) message space $\mathcal{M}$ with $|\mathcal{M}| > 1$
 - The key-generation algorithm Gen is a probabilistic algorithm that outputs a key k chosen according to some distribution. We denote by $\mathcal{K}$ the (finite) key space, i.e., the set of all possible keys that can be output by Gen .
 - The encryption algorithm Enc takes as input a key $k \in \mathcal{K}$ and a message $m \in \mathcal{M}$, and outputs a ciphertext c .
 - Let $\mathcal{C}$ denote the set of all possible ciphertexts that can be output by $\text{Enc}_k(m)$, for all possible choices of $k \in \mathcal{K}$ and $m \in \mathcal{M}$
 - The decryption algorithm Dec takes as input a key $k \in \mathcal{K}$ and a ciphertext $c \in \mathcal{C}$ and outputs a message $m \in \mathcal{M}$
 - **Perfect Correctness:**
 - We assume perfect correctness meaning that for all $k \in \mathcal{K}$, $m \in \mathcal{M}$ and any ciphertext c output by $\text{Enc}_k(m)$, it holds that $\text{Dec}_k(c) = m$ with probability 1.
 - It implies that we may assume Dec is **deterministic** without loss of generality, since $\text{Dec}_k(c)$ must give the same output every time it is run
- write $m := \text{Dec}_k(c)$

Definitions

- Let K be a random variable denoting the value of the key output by Gen
→ for any $k \in \mathcal{K}$, $\Pr[K = k]$ denotes the probability that the key output by Gen is equal to k .
- Let M be a random variable denoting the message being encrypted
→ $\Pr[M = m]$ denotes the probability that the message takes on the value $m \in \mathcal{M}$
- The *probability distribution of the message is not determined by the encryption scheme* itself, but instead reflects the likelihood of different messages being sent by the parties using the scheme, as well as an adversary's uncertainty about what will be sent.
- Example: An adversary may know that the message will either be “attack today” or “don't attack”. The adversary may even know (by other means) that with probability 0.7 the message will be a command to attack and with probability 0.3 the message will be a command not to attack:

$$\Pr[M = \text{attack today}] = 0.7 \text{ and } \Pr[M = \text{don't attack}] = 0.3$$

Definitions

- K and M are assumed to be **independent**, i.e., what is being communicated by the parties is independent of the key they happen to share.
- This makes sense as:
 - The distribution over $\mathcal{K}$ is determined by the encryption scheme itself (since it is defined by Gen)
 - While the distribution over $\mathcal{M}$ depends on the context in which the encryption scheme is being used.
- Let C be the random variable denoting the resulting ciphertext
 - for $c \in \mathcal{C}$, write $\Pr[C = c]$ to denote the probability that the ciphertext is equal to the fixed value c .

Examples

- Example for Caesar Cipher:

by definition, we have $\mathcal{K} = \{0, \dots, 25\}$ with $\Pr[K = k] = 1/26$ for each $k \in \mathcal{K}$.
Say we are given the following distribution over $\mathcal{M}$:

$$\Pr[M = \mathbf{a}] = 0.7 \quad \text{and} \quad \Pr[M = \mathbf{z}] = 0.3.$$

What is the probability that the ciphertext is **B**? There are only two ways this can occur: either $M = \mathbf{a}$ and $K = 1$, or $M = \mathbf{z}$ and $K = 2$. By independence of M and K , we have

$$\begin{aligned} \Pr[M = \mathbf{a} \wedge K = 1] &= \Pr[M = \mathbf{a}] \cdot \Pr[K = 1] \\ &= 0.7 \cdot \left(\frac{1}{26}\right). \end{aligned}$$

Similarly, $\Pr[M = \mathbf{z} \wedge K = 2] = 0.3 \cdot \left(\frac{1}{26}\right)$. Therefore,

$$\begin{aligned} \Pr[C = \mathbf{B}] &= \Pr[M = \mathbf{a} \wedge K = 1] + \Pr[M = \mathbf{z} \wedge K = 2] \\ &= 0.7 \cdot \left(\frac{1}{26}\right) + 0.3 \cdot \left(\frac{1}{26}\right) = 1/26. \end{aligned}$$

Examples

- Example for Caesar Cipher: (continued)

We can calculate conditional probabilities as well. For example, what is the probability that the message a was encrypted, given that we observe ciphertext B ? → Using Bayes' Theorem

$$\begin{aligned}\Pr[M = a \mid C = B] &= \frac{\Pr[C = B \mid M = a] \cdot \Pr[M = a]}{\Pr[C = B]} \\ &= \frac{0.7 \cdot \Pr[C = B \mid M = a]}{1/26}.\end{aligned}$$

Note that $\Pr[C = B \mid M = a] = 1/26$, since if $M = a$ then the only way $C = B$ can occur is if $K = 1$ (which occurs with probability $1/26$). We conclude that $\Pr[M = a \mid C = B] = 0.7$. $\diamond$

Examples

- Another Example for Caesar Cipher:

$$\Pr[M = \text{kim}] = 0.5, \Pr[M = \text{ann}] = 0.2, \Pr[M = \text{boo}] = 0.3.$$

What is the probability that $C = \text{DQQ}$? The only way this ciphertext can occur is if $M = \text{ann}$ and $K = 3$, or $M = \text{boo}$ and $K = 2$, which happens with probability $0.2 \cdot 1/26 + 0.3 \cdot 1/26 = 1/52$.

So what is the probability that **ann** was encrypted, conditioned on observing the ciphertext **DQQ**? A calculation as above using Bayes' Theorem gives $\Pr[M = \text{ann} \mid C = \text{DQQ}] = 0.4$. $\diamond$

Perfect Secrecy

- We imagine an adversary who knows:
 - The probability distribution over $\mathcal{M}$
 - The encryption scheme being used
- The only thing unknown to the adversary is the **key** shared by the parties.
- A message is chosen by one of the honest parties and encrypted, and the resulting ciphertext transmitted to the other party. The adversary can eavesdrop on the parties' communication, and thus observe this ciphertext → “**ciphertext-only attack**”
- For a scheme to be perfectly secret, observing this ciphertext should have no effect on the adversary's knowledge regarding the actual message that was sent:
→ the a *posteriori* probability that some message $m \in \mathcal{M}$ sent, conditioned on the ciphertext that was observed, should be no different from the a *priori* probability that m would be sent.
 - This means that the ciphertext reveals nothing about the underlying plaintext, and the adversary learns absolutely nothing about the plaintext that was encrypted.

Perfect Secrecy

- Formally:

DEFINITION An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly secret if for every probability distribution over $\mathcal{M}$, every message $m \in \mathcal{M}$, and every ciphertext $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$:

$$\Pr[M = m \mid C = c] = \Pr[M = m].$$

(The requirement that $\Pr[C = c] > 0$ is a technical one needed to prevent conditioning on a zero-probability event.)

We now give an equivalent formulation of perfect secrecy. Informally, this formulation requires that the probability distribution of the ciphertext does not depend on the plaintext, i.e., for any two messages $m, m' \in \mathcal{M}$ the distribution of the ciphertext when m is encrypted should be identical to the distribution of the ciphertext when m' is encrypted. Formally, for every $m, m' \in \mathcal{M}$, and every $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c] \quad (\text{Equation S})$$

(where the probabilities are over choice of K and any randomness of Enc)

LEMMA An encryption scheme $(\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly secret if and only if Equation S holds for every $m, m' \in \mathcal{M}$ and every $c \in \mathcal{C}$.

This implies that the ciphertext contains no information about the plaintext, and that it is impossible to distinguish an encryption of m from an encryption of m' since the distributions over the ciphertext are the same in each case.

Perfect Secrecy

- Proof: we show that if $\Pr[\text{Enc}_K(m) = c] = \Pr[\text{Enc}_K(m') = c]$ holds, then the scheme is perfectly secret:

Fix a distribution over $\mathcal{M}$, a message m , and a ciphertext c for which $\Pr[C = c] > 0$. If $\Pr[M = m] = 0$ then we trivially have

$$\Pr[M = m \mid C = c] = 0 = \Pr[M = m].$$

So, assume $\Pr[M = m] > 0$. Notice first that

$$\Pr[C = c \mid M = m] = \Pr[\text{Enc}_K(M) = c \mid M = m] = \Pr[\text{Enc}_K(m) = c],$$

where the first equality is by definition of the random variable C , and the second is because we condition on the event that M is equal to m . Set $\delta_c \stackrel{\text{def}}{=} \Pr[\text{Enc}_K(m) = c] = \Pr[C = c \mid M = m]$. If the condition of the lemma holds, then for *every* $m' \in \mathcal{M}$ we have $\Pr[\text{Enc}_K(m') = c] = \Pr[C = c \mid M = m'] = \delta_c$. Using Bayes' Theorem we thus have

$$\begin{aligned} \Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\sum_{m' \in \mathcal{M}} \Pr[C = c \mid M = m'] \cdot \Pr[M = m']} \\ &= \frac{\delta_c \cdot \Pr[M = m]}{\sum_{m' \in \mathcal{M}} \delta_c \cdot \Pr[M = m']} \\ &= \frac{\Pr[M = m]}{\sum_{m' \in \mathcal{M}} \Pr[M = m']} = \Pr[M = m], \end{aligned}$$

where the summation is over $m' \in \mathcal{M}$ with $\Pr[M = m'] \neq 0$. We conclude that for every $m \in \mathcal{M}$ and $c \in \mathcal{C}$ for which $\Pr[C = c] > 0$, it holds that $\Pr[M = m \mid C = c] = \Pr[M = m]$, and so the scheme is perfectly secret. ■


$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

Perfect (Adversarial) Indistinguishability

- Another equivalent definition of perfect secrecy.
- This definition is based on an **experiment** involving an adversary passively observing a ciphertext and then trying to guess which of two possible messages was encrypted.
- We consider the following experiment:

The adversarial indistinguishability experiment $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}}$:

1. The adversary $\mathcal{A}$ outputs a pair of messages $m_0, m_1 \in \mathcal{M}$.
2. A key k is generated using Gen , and a uniform bit $b \in \{0, 1\}$ is chosen. Ciphertext $c \leftarrow \text{Enc}_k(m_b)$ is computed and given to $\mathcal{A}$. We refer to c as the challenge ciphertext.
3. $\mathcal{A}$ outputs a bit b' .
4. The output of the experiment is defined to be 1 if $b' = b$, and 0 otherwise. We write $\text{PrivK}_{\mathcal{A},\Pi}^{\text{eav}} = 1$ if the output of the experiment is 1 and in this case we say that $\mathcal{A}$ succeeds.

$\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ be an encryption scheme

One of these two messages is chosen **uniformly at random** and encrypted using a random key

it is trivial for $\mathcal{A}$ to succeed with probability $1/2$ by outputting a random guess. Perfect indistinguishability requires that it is impossible for any $\mathcal{A}$ to do better.

Perfect (Adversarial) Indistinguishability

DEFINITION *Encryption scheme $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ with message space $\mathcal{M}$ is perfectly indistinguishable if for every $\mathcal{A}$ it holds that*

$$\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = \frac{1}{2}.$$

LEMMA *Encryption scheme Π is perfectly secret if and only if it is perfectly indistinguishable.*

Example:

We show that the Vigenère cipher is **not** perfectly indistinguishable, at least for certain parameters. Concretely, let Π denote the Vigenère cipher for the message space of two-character strings, and where the **period** is chosen uniformly in $\{1, 2\}$. To show that Π is not perfectly indistinguishable, we exhibit an adversary $\mathcal{A}$ for which $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] > \frac{1}{2}$.

the length of the key

Adversary $\mathcal{A}$ does:

1. Output $m_0 = \text{aa}$ and $m_1 = \text{ab}$.
2. Upon receiving the challenge ciphertext $c = c_1 c_2$, do the following: if $c_1 = c_2$ output 0; else output 1.

Computation of $\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1]$ is tedious but straightforward.

$$\begin{aligned} \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] &= \frac{1}{2} \cdot \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1 \mid b = 0] + \frac{1}{2} \cdot \Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1 \mid b = 1] \\ &= \frac{1}{2} \cdot \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0] + \frac{1}{2} \cdot \Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1], \end{aligned} \quad (2.2)$$

where b is the uniform bit determining which message gets encrypted. $\mathcal{A}$ outputs 0 if and only if the two characters of the ciphertext $c = c_1 c_2$ are equal. When $b = 0$ (so $m_0 = \text{aa}$ is encrypted) then $c_1 = c_2$ if either (1) a key of period 1 is chosen, or (2) a key of period 2 is chosen, and both characters of the key are equal. The former occurs with probability $\frac{1}{2}$, and the latter occurs with probability $\frac{1}{2} \cdot \frac{1}{26}$. So

Why??

$$\Pr[\mathcal{A} \text{ outputs } 0 \mid b = 0] = \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} \approx 0.52.$$

When $b = 1$ then $c_1 = c_2$ only if a key of period 2 is chosen and the first character of the key is one more than the second character of the key, which happens with probability $\frac{1}{2} \cdot \frac{1}{26}$. So

$$\Pr[\mathcal{A} \text{ outputs } 1 \mid b = 1] = 1 - \Pr[\mathcal{A} \text{ outputs } 0 \mid b = 1] = 1 - \frac{1}{2} \cdot \frac{1}{26} \approx 0.98.$$

Plugging into Equation (2.2) then gives

$$\Pr [\text{PrivK}_{\mathcal{A}, \Pi}^{\text{eav}} = 1] = \frac{1}{2} \cdot \left(\frac{1}{2} + \frac{1}{2} \cdot \frac{1}{26} + 1 - \frac{1}{2} \cdot \frac{1}{26} \right) = 0.75 > \frac{1}{2},$$

and the scheme is not perfectly indistinguishable. $\diamond$

The One-Time Pad

- Using *binary and XOR as Vernam* and use only the *key one time as One-Time Pad*.

Fix an integer $\ell > 0$. The message space $\mathcal{M}$, key space $\mathcal{K}$, and ciphertext space $\mathcal{C}$ are all equal to $\{0, 1\}^\ell$ (the set of all binary strings of length ℓ).

- Gen: the key-generation algorithm chooses a key from $\mathcal{K} = \{0, 1\}^\ell$ according to the uniform distribution (i.e., each of the 2^ℓ strings in the space is chosen as the key with probability exactly $2^{-\ell}$).
- Enc: given a key $k \in \{0, 1\}^\ell$ and a message $m \in \{0, 1\}^\ell$, the encryption algorithm outputs the ciphertext $c := k \oplus m$.
- Dec: given a key $k \in \{0, 1\}^\ell$ and a ciphertext $c \in \{0, 1\}^\ell$, the decryption algorithm outputs the message $m := k \oplus c$.

THEOREM

The one-time pad encryption scheme is perfectly secret.

The One-Time Pad

PROOF We first compute $\Pr[C = c \mid M = m']$ for arbitrary $c \in \mathcal{C}$ and $m' \in \mathcal{M}$. For the one-time pad,

$$\begin{aligned}\Pr[C = c \mid M = m'] &= \Pr[\text{Enc}_K(m') = c] = \Pr[m' \oplus K = c] \\ &= \Pr[K = m' \oplus c] \\ &= 2^{-\ell},\end{aligned}$$

where the final equality holds because the key K is a uniform ℓ -bit string. Fix any distribution over $\mathcal{M}$. For any $c \in \mathcal{C}$, we have

$$\begin{aligned}\Pr[C = c] &= \sum_{m' \in \mathcal{M}} \Pr[C = c \mid M = m'] \cdot \Pr[M = m'] \\ &= 2^{-\ell} \cdot \sum_{m' \in \mathcal{M}} \Pr[M = m'] \\ &= 2^{-\ell},\end{aligned}$$

where the sum is over $m' \in \mathcal{M}$ with $\Pr[M = m'] \neq 0$. Bayes' Theorem gives:

$$\begin{aligned}\Pr[M = m \mid C = c] &= \frac{\Pr[C = c \mid M = m] \cdot \Pr[M = m]}{\Pr[C = c]} \\ &= \frac{2^{-\ell} \cdot \Pr[M = m]}{2^{-\ell}} \\ &= \Pr[M = m].\end{aligned}$$

We conclude that the one-time pad is perfectly secret. ■

The One-Time Pad

- One-time pad encryption is **rarely used** any more due to a number of drawbacks:
 - Most prominent is that the **key is as long as the message**. This limits the usefulness of the scheme for sending very long messages (as it may be difficult to securely share and store a very long key), and is problematic when *the parties cannot predict in advance (an upper bound on) how long the message will be*.
 - The one-time pad is only secure if used **once (with the same key)**. It is easy to see that encrypting more than one message with the same key leaks a lot of information:

Two messages m and m' are encrypted using the same key k .
An adversary who obtains c and c' can compute:

$$c \oplus c' = (m \oplus k) \oplus (m' \oplus k) = m \oplus m'$$

Thus, learn the XOR of the two messages or, equivalently, exactly where the **two messages differ**. While this may not seem very significant, it is enough to rule out any claims of perfect secrecy for encrypting two messages using the same key.

Moreover, if the messages correspond to natural-language text, then given the XOR of two sufficiently long messages it is possible to perform frequency analysis and recover the messages themselves.

Limitations of Perfect Secrecy

- The drawbacks mentioned for the one-time pad encryption scheme are not specific to that scheme, but are instead **inherent limitations of perfect secrecy**.
- Specifically, we prove that any perfectly secret encryption scheme must have a *key space that is at least as large as the message space*.

THEOREM If $(\text{Gen}, \text{Enc}, \text{Dec})$ is a perfectly secret encryption scheme with message space $\mathcal{M}$ and key space $\mathcal{K}$, then $|\mathcal{K}| \geq |\mathcal{M}|$.

If all keys are the same length, and the message space consists of all strings of some fixed length

→ this implies that the **key is at least as long as the message**.

In particular, the key length of the one-time pad is optimal.

Limitations of Perfect Secrecy

PROOF We show that if $|\mathcal{K}| < |\mathcal{M}|$ then the scheme cannot be perfectly secret. Assume $|\mathcal{K}| < |\mathcal{M}|$. Consider the uniform distribution over $\mathcal{M}$ and let $c \in \mathcal{C}$ be a ciphertext that occurs with non-zero probability. Let $\mathcal{M}(c)$ be the set of all possible messages that are possible decryptions of c ; that is

Note: c is constant, m and k changes.

$$\mathcal{M}(c) \stackrel{\text{def}}{=} \{m \mid m = \text{Dec}_k(c) \text{ for some } k \in \mathcal{K}\}.$$

Clearly $|\mathcal{M}(c)| \leq |\mathcal{K}|$. (Recall that we may assume Dec is deterministic.) If $|\mathcal{K}| < |\mathcal{M}|$, there is some $m' \in \mathcal{M}$ such that $m' \notin \mathcal{M}(c)$. But then

$$\Pr[M = m' \mid C = c] = 0 \neq \Pr[M = m'],$$

and so the scheme is not perfectly secret. ■

The above theorem shows an inherent limitation of schemes that achieve perfect secrecy. Even so, individuals occasionally claim they have developed a new encryption scheme that is “unbreakable” and achieves the security of the one-time pad without using keys as long as what is being encrypted. The above proof demonstrates that **such claims cannot be true.**



Thank You